

is a positive infinitesimal.) This shows incidentally that Δ_F is a Green's function for the Klein-Gordon differential operator, in the sense that

$$(\square - m^2)\Delta_F(x) = -\delta^4(x) \quad (6.2.17)$$

with boundary conditions specified by the $-i\epsilon$ in the denominator: as shown by Eq. (6.2.13), $\Delta_F(x)$ for $x^0 \rightarrow +\infty$ or $x^0 \rightarrow -\infty$ involves only positive or negative frequency terms, $\exp(-ix^0\sqrt{\mathbf{p}^2 + m^2})$ or $\exp(+ix^0\sqrt{\mathbf{p}^2 + m^2})$, respectively.

Inserting Eq. (6.2.16) in Eq. (6.2.14) now gives the propagator as

$$\Delta_{\ell m}(x, y) = (2\pi)^{-4} \int d^4q \frac{P_{\ell m}^{(L)}(q)e^{iq \cdot (x-y)}}{q^2 + m^2 - i\epsilon}. \quad (6.2.18)$$

There is one obvious problem with this expression. The polynomial $P(p)$ is Lorentz-covariant when p is on the mass shell, $p^2 = -m^2$, but in Eq. (6.2.18) we integrate over all q^μ , not restricted to the mass shell. The polynomial $P^{(L)}(q)$ is defined for general q^μ to be linear in q^0 , a condition that clearly does not respect Lorentz covariance unless the polynomial is also linear in each spatial component q^i as well. We can instead always define our extension of the polynomial $P(p)$ to general four-momenta q^μ , which we shall call simply $P(q)$, in such a way that $P(q)$ is Lorentz-covariant for general q^μ , in the sense that

$$P_{\ell m}(\Lambda q) = D_{\ell \ell'}(\Lambda) D_{mm'}^*(\Lambda) P_{\ell' m'}(q),$$

where Λ^μ_ν is a general Lorentz transformation, and $D(\Lambda)$ is the appropriate representation of the Lorentz group. For instance, for scalar, Dirac, and four-vector fields, these covariant extensions are obviously provided by just replacing p^μ with a general four-momentum q^μ in Eqs. (6.2.4), (6.2.5), and (6.2.6). For the scalar and Dirac fields, these are already linear in q^0 , so here there is no difference between $P^{(L)}(q)$ and $P(q)$:

$$P_{\ell m}^{(L)}(q) = P_{\ell m}(q) \quad (\text{scalar, Dirac fields}). \quad (6.2.19)$$

On the other hand, for the vector field of a spin one particle, the 00 components of the covariant polynomial $P_{\mu\nu}(q) \equiv \eta_{\mu\nu} + m^{-2}q_\mu q_\nu$ are quadratic in q^0 , so here there is a difference:

$$\begin{aligned} P_{\mu\nu}^{(L)}(q) &= \eta_{\mu\nu} + m^{-2} [q_\mu q_\nu - \delta_\mu^0 \delta_\nu^0 (q_0^2 - \mathbf{q}^2 - m^2)] \\ &= P_{\mu\nu}(q) + m^{-2} (q^2 + m^2) \delta_\mu^0 \delta_\nu^0. \end{aligned} \quad (6.2.20)$$

(The extra term here is fixed by the two conditions that it must cancel the $(q_0)^2$ term in $P_{00}(q)$, and must vanish when q^μ is on the mass shell.) Inserting this in Eq. (6.2.18) gives the propagator of a vector field as

$$\Delta_{\mu\nu}(q) = (2\pi)^{-4} \int d^4q \frac{P_{\mu\nu}(q)e^{iq \cdot (x-y)}}{q^2 + m^2 - i\epsilon} + m^{-2} \delta^4(x-y) \delta_\mu^0 \delta_\nu^0. \quad (6.2.21)$$